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Binding sex, age, and race in unfamiliar faces: The formation of "face files"

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ABSTRACT

A series of five experiments investigated the binding of social categories (e.g., sex, race, and age) and non-facial attributes (e.g., spatial location and response) in unfamiliar faces. Evidence for the creation and retrieval of temporary, short-term memory structures, across perception and action has been adduced. The binary bindings documented here (e.g., sex and race, race and spatial location) were indicated by *partial repetition costs*, in which repeating a pair of social categories or altering them altogether led to faster responses than repeating or alternating only one of them. These episodic structures, dubbed herein "face files", consisted of both visuo–visuo and visuo–motor integrations. Their presence suggests that sets of binary social categories in unfamiliar faces are extracted simultaneously and retrieved together automatically. The implications of these results for theories of person construal and social cognition are discussed.

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1. Introduction

Faces provide us with a great deal of information regarding their owners. This information includes, but is not confined, to such social categories as sex, age, or race (Bruce & Young, 1986; Fitousi & Wenger, 2013; Fiske & Neuberg, 1990; Freeman & Ambady, 2011). Extracting social facial attributes from faces is held most efficiently and rapidly by all people. However, this efficiency conceals the fact that complex processes are involved (Allport, 1954; Bargh, 1999; Brewer, 1988; Fiske & Neuberg, 1990). Researchers in the domains of cognitive psychology (Algom & Fitousi, 2016; Bruce & Young, 1986; Burton, Bruce, & Johnston, 1990; Calder & Young, 2005; Farah, Wilson, Drain, & Tanaka, 1998; Fitousi & Wenger, 2013; Fitousi, 2013, 2015, 2016a; Haxby, Hoffman, & Gobbini, 2000) and social psychology (Fiske & Neuberg, 1990; Freeman & Ambady, 2011; Levin, 2000; Macrae, Bodenhausen, & Milne, 1995; Quinn & Macrae, 2005) have invested great efforts in studying the psychological mechanisms involved in the extraction of facial attributes.

The current study presents evidence for the existence of a binding mechanism by which binary combinations of social categories in unfamiliar faces such age and race, or age and sex are encoded and retrieved together from short-term memory in a rapid an involuntary fashion. The proposed mechanism is investigated here using a priming method (Hommel, 1998), and is conceived within a novel theoretical framework called "face files" (Fitousi, 2016b). The notion of face files bears on earlier classic ideas and methodologies advanced in the visual attention literature (Kahneman, Treisman, & Gibbs, 1992; Hommel, 1998; Kahneman & Treisman, 1984). The finding of face files supports the idea that social categories of unfamiliar faces are extracted simultaneously and automatically (Bargh, 1999). The results have interesting implications for theories of person construal, face processing, and social cognition.

1.1. The extraction of social categories

How are social aspects of unfamiliar faces extracted? This question has been central to modern accounts of "person construal" (Brewer, 1988; Fiske & Neuberg, 1990; Freeman & Ambady, 2011). According to a common idea in the study of person construal, thinking is categorical (Allport, 1954). This means that the mere exposure to a social stimulus, such as a human face, elicits high level constructs (e.g., MAN, OLD). These representations are abstracted from the low-level features (e.g., skin tone, eye shape) that constitute a face (Bargh, 1999; Devine, 1989). In turn, these representations trigger associated behaviors and attitudes (e.g., stereotypes) toward the perceived person. According to a related idea, when perceiving a face, not only the focal category is activated (e.g., sex), but rather a whole set of other social attributes (e.g., age, race) are competing for attention (Freeman & Ambady, 2011). Evidence for this notion comes from various sources. Quinn and Macrae (2005) have shown, using a repetition priming paradigm, that participants who engaged in a sex categorization task with faces responded more rapidly if, at an earlier stage of the experiment they had performed an age rather than sex categorization task with the same faces. This apparent sensitivity to task-irrelevant category has been found also in an application of the Garner paradigm (Garner, 1974; for a review see Algom & Fitousi, 2016) to these social categories. Participants who were performing in a sex categorization task were

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interfered by irrelevant variations on age, such that performance was worse in blocks in which faces varied on age (i.e., filtering) compared to blocks in which age was held constant (i.e., baseline). Interestingly, this Garner interference was absent when participants were judging age and ignoring variations on sex.

Another source of evidence for categorical perception that is shaped by multiple categories activation has been demonstrated in a mouse tracking paradigm (Cloutier, Freeman, & Ambady, 2014; Freeman, Nakayama, & Ambady, 2013). In these studies, participants classify a relevant facial attribute (e.g., sex) by moving their mouse toward the correct label (e.g., FEMALE) on a computer screen. Participants should ignore an irrelevant facial attribute (e.g., age) and its pertinent label. The common finding in the mouse-tracking paradigm is that participants' trajectories toward the relevant social category label are biased by the irrelevant category. These results have been taken as strong evidence that social categories are activated simultaneously while competing for response, even when they are not relevant for the task at hand. One interesting observation that is common to the mouse-tracking paradigm (Freeman et al., 2013), as well as to the repetition priming (Quinn & Macrae, 2005), and Garner (Quinn & Macrae, 2005) results concerns the possibility that social categories elicits activations patterns that are not necessarily symmetric. For example, Cloutier et al. (2014) found that age affected classification trajectories of sex, but sex did not affect the trajectories of age categorizations.

The evidence adduced for the activation of multiple social categories in faces raises a related question. How are multiple social categories combined in faces to provide us with the united phenomenal experience they often elicit? In spite of its theoretical importance, rarely has this question been addressed in the person construal literature.

1.2. How are social categories integrated?

Imagine you are introduced to the two (fictitious) characters of Tom and Sara. It is both useful and customary to garner as much information from their faces in order to learn something new about their background, origin, or social status. This information might prove essential for your survival (e.g., mating, escape). Your visual system, thus, embarks on the mission of extracting Tom's and Sara's sex (Man, Woman), age (Old, Young), race (Asian, White), as well as other facial attributes (e.g., facial expression, gaze-direction) from featural aspects of their face. Yet, a crucial problem may challenge your visual system. The facial codes extracted (Male, Female, White, Young) should be attached to the pertinent persons in their correct configuration (e.g., "Sara + Woman + Young" vs. "Tom + Man + Black"). Success in such a task, by all means, is not trivial. This is mainly because faces are subjected to dynamic changes with respect to their featural and spatial aspects (e.g., change of expression, lighting, or location in space). Hence, perceivers must prevent situations in which codes that belong to different people are mixed up. Potential difficulties in extraction and binding of social categories can be considered as a part of a more general problem of the human information processer called "the binding problem" (Singer & Gray, 1995; Treisman, 1996; von der Malsburg, 1999). The next section provides a brief overview of the problem.

1.3. The binding problem

A yellow banana is perceived as a single object with its shape and color being part of a unitary whole. When this yellow banana is presented next to a red apple, observers, most often, would perceive the colors and shapes of the two objects in their right configuration. That is, "yellow banana + red apple." Observers do not perceive the red color of the apple as belonging to the banana, nor do they experience the yellow color of the banana as belonging to the apple. Perceiving these objects in their correct configuration is a fairly sophisticated process, reflecting a non-trivial ability of the primate brain. The reason is that the elementary features that compose perceptual objects are represented: (a) categorically (Freedman, Riesenhuber, Poggio, & Miller, 2001), and (b) in a distributed manner (Hubel & Wiesel, 1977; Felleman & Van Essen, 1991). That is, the basic attributes of color, shape, or location are coded as perceptual types, and are represented as neuronal codes across separate brain areas in the visual cortex (Livingstone & Hubel, 1987, 1988). Recombining these attributes' codes into the veridical representations of the viewed objects poses a major computational challenge to our perceptual systems (Treisman & Gelade, 1980). This challenge is known as the *binding problem* (Singer & Gray, 1995; Treisman, 1996; von der Malsburg, 1999).

Extensive effort has been dedicated to the understanding of the binding problem over the last thirty years or so in the field of attention (Kahneman et al., 1992). Two major experimental approaches have been broadly used. One method relies on the investigation of "illusory conjunctions" (Treisman & Schmidt, 1982) – cases in which observers erroneously report features from one object as belonging to its spatial neighbor. A second method is based on the observation of aftereffects generated by featural information (Kahneman et al., 1992; Hommel, 1998). Kahneman, Treisman, and their colleagues have performed a systematic analysis of feature binding using a priming task (Kahneman & Treisman, 1984; Kahneman et al., 1992). In this task, a letter appeared in a prime display, and then the same letter or different letter was presented in a probe display. The results showed that the probe letter was named faster if the letter's identity was repeated and associated with the same object/location. Kahneman et al. (1992) argued that processing of a visual object leads to the creation of an "object file" – an episodic representation of the object's identity and location – allowing its identification in spite of spatiotemporal discontinuities.

Hommel (1998) has advanced our understanding of the binding processes at both methodological and theoretical levels. Using a similar priming paradigm as that of Kahneman et al. (1992), he showed, that repeating two given features (e.g., a red square) or alternating them altogether (e.g., a blue triangle) gave rise to improved performance levels relative to conditions in which only one of the features repeated and the other alternated (e.g., a red triangle). According to Hommel, this interaction pattern represents a processing cost, rather than a benefit (Hommel & Colzato, 2009), and is therefore called *partial-repetition cost* (Hommel, 2004, p. 496). This pattern entails that "objects files" consist of a subset (i.e., binary bindings) of their features, not the entire list of features suggested by Kahneman et al. (1992). In addition, Hommel introduced the concept of "*event file*" – a temporary binding in which a response is integrated with a perceptual feature, resulting in a visuo-motor episodic trace (Hommel, 1998). Thus, responding to a green object with your left hand may create a binding structure that consists of the green color with the motor code associated with your left hand.

1.4. Binding problems with faces

Work on the binding problem has been often confined to simple colored objects (but see Keizer, Colzato, & Hommel, 2008.; Fitousi, 2016b), although more complex objects such as faces may be subjected to similar or even more sophisticated binding processes. The potential role of binding in face recognition is supported by several lines of evidence. First, the need for binding can be inferred from cases in which it fails. In the well-known McGurk effect (McGurk & MacDonald, 1976), the vocal sound produced by a face is erroneously integrated with its lips' movements such that the perceiver hears a different phoneme than that articulated. Similarly, initial primitive social categorizations are slowed down or become inaccurate when additional incongruent cues are presented with the face. Sex classifications are hampered when hairstyle or hair length are incongruent with the sex of the face (Brebner, Martin, & Macrae, 2009; Burton, Bruce, & Dench, 1993). Race judgments are slowed down when skin tone and shape of individual face features are incongruent with the race of the face (Enlow &

Moyers, 1982; Levin, 2000; Maclin & Malpass, 2001). Third, there is evidence for both the categorical perception of facial attributes of social import (Bestelmeyer et al., 2008; Etcoff & Magee, 1992), as well as for their distributed brain coding (Breiter et al., 1996; Haxby et al., 2000; Kanwisher, McDermott, & Chun, 1997; Rogers & McClelland, 2004; Spivey & Dale, 2004; Rolls & Tovee, 1995; Sugase, Yamane, Ueno, & Kawano, 1999). These two aspects of face processing entail that the binding problem may apply to faces as much as it applies to simple objects.

1.5. The formation of "face files"

Most researchers would agree that faces are more complex entities than simple objects (e.g., a red square). One reason is that although faces and objects are perceived through multiple sensory modalities, only with faces do these sources convey valuable information. A second reason is that face recognition relies on various types of cues. As a result, binding processes with faces require a different conceptual framework than that conceived for objects. Fitousi (2016b) has recently proposed such a framework. The "face files" approach extends previous ideas on "object files" and "event files" (Kahneman et al., 1992; Hommel, 1998) into a new account of face recognition. The core idea behind "face files" is that people represent faces in short-term memory by creating, maintaining, and retrieving sets of binary episodic conjunctions of facial and non-facial features. These structures are formed automatically with the mere presentation of faces. The pairwise conjunctions of facial attributes may consist of such crucial attributes as facial identity and emotion; but they may also include pairwise conjunctions of a facial attribute and the motor action emitted toward the face (e.g., binding of facial emotion and response), or a binding of the face's spatial location with one of its social categories (e.g., location and gaze direction).

Fitousi (2016b) has recently deployed the method of Hommel (1998) to investigate the binding of facial identity, emotion, gaze-direction, spatial location, and response location. On each trial, participants were presented with a prime face S1 (e.g., Tom + Sad), followed by an arrow cue (press left, or press right) for response, and then a target face S2 (e.g., Tom + Happy). Participants had to respond to the face in S1 according to direction of the cue – R1 – and then to the face in S2 according to the task-relevant facial attribute – R2 (e.g., sad, happy). Binding effects were indicated by *partial repetition costs* – faster RTs in cases where two attributes repeated or alternated together compared to cases in which only one of the two features repeated and the other alternated. The patterns observed were explained in terms of "face files", temporary, binary integrations of facial and non-facial features. Fitousi (2016b) found that the task-relevant facial dimension was always bound with the motor response (i.e., event file). Second, facial identity (e.g., Tom) was integrated automatically into a "face file", whether or not it served as the task-relevant dimension (see for Fitousi, 2016b, Experiment 1 and 2). Third, the integration of spatial location into a "face file" depended on the spatial nature of the other facial attribute to which it was bound. For example, gaze-direction, which is coded spatially, was bound with spatial location, whereas facial identity and emotion, which most likely are not coded spatially, were not integrated with location. Fourth, there were asymmetries in binding. Gaze-direction was bound with emotion, but emotion was not bound with gaze-direction. All of these might suggest the presence of "face files." Face files can provide a valuable framework for understanding the binding of social categories such as sex, race, and age.

1.6. The Martin et al. (2015) study

Recently, Martin et al. (2015) have reported a study in which participants engaged in a sequential classification task of the social categories of sex, age, or race. On each experiment participants were asked to respond to a relevant social category (e.g., sex) while ignoring trial-to-trial variations on other irrelevant social categories (e.g., race, age). Martin et al. found *partial repetition costs* from trial *n-1* to trial *n*, such that performance was improved when the relevant and irrelevant social categories repeated or alternated together compared to trials in which only one of the categories repeated and the other alternated. The method harnessed by Martin et al. bears close affinity with the current priming setup (see also Fitousi, 2016b). However, the present study differs from the study of Martin et al. (2015) in several fundamental ways. First, borrowed from the language switching literature (Thomas & Allport, 2000), the method used by Martin et al. was not initially developed to study binding effects, and as such is lacking the theoretical underpinnings that afford principled investigation of the binding process. For example, the Martin el al method cannot, in principle, separate response binding from social category binding. Assume that S1 (i.e., Face 1) and S2 (i.e., Face 2) repeat with respect to their sex. It is not clear whether a speed up in RTs is due to repetition of the response to S2 (R2 = R1) or due to the repetition of the social category itself (S1 = S2), and ditto to cases of complete or partial alternations (for a similar argument see Hommel, Proctor, and Lu, 2004).The current method has been specifically engineered to disentangle stimulus and response effects (Hommel, 1998, 2004).

A second problem in the study of Martin et al. concerns the fact that the social category of race in their study was not well controlled, as faces were either white or not-white. So it is not clear whether participants in their study actually extracted the race of the faces. Third, the sequential paradigm used by Martin et al. does not afford a balanced design with an equal number of single, pairwise, and triple repetitions and alternations of all social categories within a block. This imbalance creates, by default correlations between the pertinent social categories (Kareev, 1995). It is well known that observers are good "correlation hunters" (Dishon-Berkovits & Algom, 2000) who can use dimensional correlation for their own advantage. This might lead participants to endogenously direct their attention to the irrelevant social category (see for Melara & Algom, 2003). Such a possibility undermines the conclusions of Martin's el al that irrelevant social categories were attended involuntarily. Finally, the potentially vital role of spatial location in binding could not be investigated in Martin et al. paradigm. Spatial location has been shown to fulfill a major (Fitousi, 2016b; Hommel, 1998, 2004) if not an indispensable (Kahneman et al., 1992) role in binding. Therefore, the methodology employed in the current study seems to be a more suitable one for studying binding effects with social categories as it offers potential remedies for all of these problems.

1.7. Research strategy

Fitousi (2016b) has recently provided substantial evidence for the existence of "face files" with the social categories of facial identity, emotion, and gaze direction. Bindings in "face files" are indicated by *partial repetition costs* (Hommel, 2004). That is, faster RTs obtained when two facial categories are repeated or alternated together compared to cases in which one of the categories is repeated and the other is alternated. The purpose of the current study has been to further test whether "face files" are created for the social categories of race, age, and sex. These dimensions have been identified by researchers as dominant categories in person perception (Brewer, 1988; Fiske & Neuberg, 1990). Five experiments were conducted to test this idea. The methods employed throughout the experiments have been developed by Hommel (1998), and extended to the realm of faces by Fitousi (2016b). These methods are applied here to the social categories of race, sex, and age. Each experiment tested binding between two of the three social categories (e.g., sex and race), with one of the categories serving as the task-relevant category and the other as an irrelevant category. In addition, response and spatial location features were also varied in order to investigate their potential role in the formation of "face files."

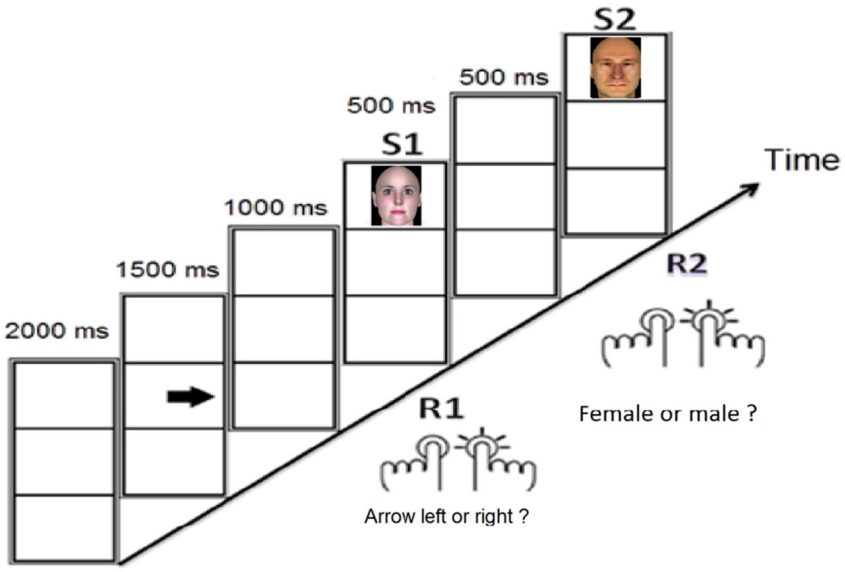


Fig. 1. An illustration of the trial sequence in Experiments 1–5. R1 and R2 are responses to the first stimulus (S1) and second stimulus (S2).

2. General method

2.1. Participants

Eighty four volunteers (70 women and 14 men) from Ariel University took part in Experiments 1–5 (mean age 26.4).[1] All of the participants were Caucasian male and female undergraduate students (aged 20–28). They participated in partial fulfillment of course credit. All reported normal or corrected-to-normal vision, normal hearing, and unencumbered use of their two hands.

2.2. Apparatus and stimuli

The experiments were controlled by a desktop computer. Viewing distance subtended 76 cm from the computer screen. The stimuli consisted of three 3.16° × 2.7° black square outlines arranged vertically from top to bottom (see Fig. 1). The faces were color images of computerized faces generated by the FaceGen software (see Fig. 2). The faces varied on their sex, race and age. Four different images were created for each experiment using a 2 × 2 factorial combination with two social categories and two levels of each category. For example, in Experiment 1, in which the binding of age and race was tested, two levels of age (young and old) were combined with two levels of race (Asian and European), to create four different faces. Each face image subtended 1.88° × 2.33°. Each face could appear either in the upper or lower box (see Fig. 1). A middle box, at the center of screen, was used for presenting the cue for response (R1). Response cues were full black arrows pointing either to the left or to the right. Responses were made by pressing the left ("z") or right ("m") keys on the keyboard.

2.3. Procedure and design

The procedure and design were similar to those reported by Hommel (1998) and Fitousi (2016b). Each experimental trial started with an arrow cue, accompanied by the presentation of a target face (S1). Participants made a response (R1) to S1 according to the cue. A leftward pointing arrow required a left-hand-key response, while a rightward pointing arrow required a right-hand-key response. Participants were informed that there would be no systematic relationship between S1 and R1, so that they should execute the precued response at the onset of S1, while ignoring the irrelevant dimension of S1. A second response (R2) was a binary-choice reaction to the second stimulus (S2). The task-relevant social category in S2 (e.g., race) was defined in advanced for each experiment. A right-hand response ("m") was assigned to one level of the task-relevant category (e.g., Asian), and a left-hand response ("z") was assigned to its second level (e.g., European).

Fig. 1 depicts a typical sequence of events in a trial. Each trial began with an arrow cue presented for 1500 ms followed by a blank interval for 500 ms. Then S1 face appeared for 500 ms and R1 was expected. S1 was then replaced by another blank interval for 500 ms followed by S2. At this stage, R2 was expected. S2 remained on the screen for 2500 ms or until response. An inter-trial interval of 2500 ms preceded the presentation of a new response cue. A block consisted of 128 trials, which were created by the factorial combination of: S1 task-relevant category (e.g., Asian, European), R1 response (left vs right), S1 task-irrelevant category (e.g., Old, Young), S2 task-relevant category (e.g., Asian, European), S2 task-irrelevant category (e.g., Old, Young), and S2 spatial location (e.g., top, bottom). The experiment consisted of three blocks of trials, amounting to a total of 384 trials per participant. The order of trials in each block was chosen randomly by the computer. A 1 min break was allowed between the blocks.

2.4. Data analysis

Trials in which RTs were incorrect, longer than 2500 ms, or shorter than 150 ms were removed from the analyses of all Experiments. These amounted to <8% in all experiments. In all five experiments, error data did not show significant effects, and thus analyses on errors were not reported. Mean RTs were calculated for each possible level of stimuli and responses in the two tasks (R1 and R2). A four-way ANOVA with the task-relevant social category (repeated, alternated), task-irrelevant social category (repeated, alternated),

[1] A power analysis has been performed using the Bausell and Li (2002) recommendations. Standard values of beta = 80% and alpha = 5% were assumed. For a two-way interaction, which is the primary theoretical concern of the current study, the required N is derived by first computing the expected differences of differences between the groups (e.g., all are the same participants). Based on previous studies (Fitousi, 2016b; Hommel, 1998) the effect size is around 100 ms. The effect size is normalized by the expected SD, which is around 10. The next step is to divide it by 2 in order to get the standardized effects size ES. In our case the ES equals to 5 (=10/2). And then looking at the Tables provided by Bausell and Li it can be seen that a total of N = 12 is an adequate number for ES of 5. Note that according to tables provided by Bausell and Li (2002), an ES of 3 requires N = 12. So even if we adopt a more conservative effect size of 60 ms we still get this N. All the studies here were conducted with an N that is equal to or >12 participants.

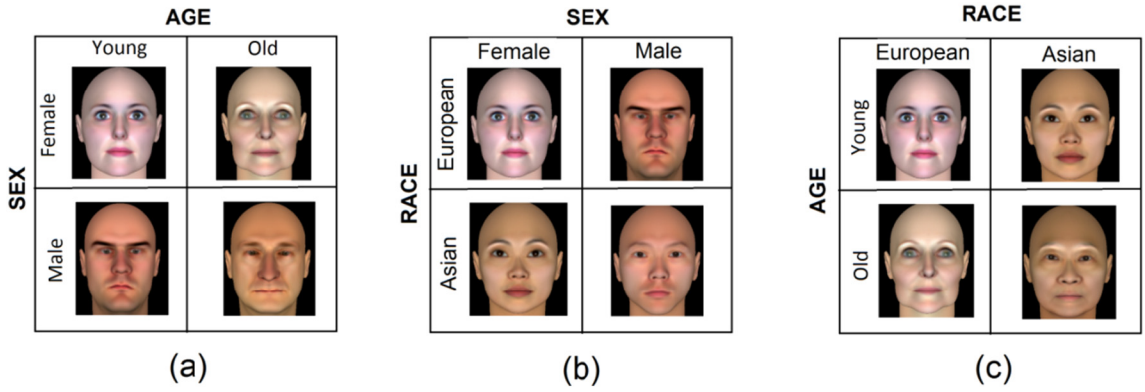


Fig. 2. An illustration of the faces displayed in Experiments 1–5. Each set of faces was constructed by a factorial combination of two facial categories: (a) age and sex (Experiments 2, 3), (b) race and sex (Experiments 4, 5), and (c) race and age (Experiment 1).

response (repeated, alternated), and the face's spatial location (repeated, alternated) as factors was performed on mean RTs. Bindings were indicated by the two-way interaction between two social categories, social category and response, or social category and spatial location. All measures, manipulations, and exclusions in the study are disclosed, as well as the method of determining the final sample size (with particular reference to whether collection was continued after data analysis).

Binding should be indicated by the following pattern (Hommel, 1998): faster mean RTs when two categories are repeated or alternated together compared to conditions in which only one of the categories is repeated and the other is alternated. It should be highlighted here that interpreting binding effects strictly requires focusing on the interaction as such (Hommel, 1998). The main effects, whether significant or insignificant, are irrelevant to the interpretation of the binding effect. Moreover, the actual effect is based on both (a) binding of the features and (b) retrieval of their combination. The presence of partial repetition cost clearly indicates that binding took place. The absence of an effect might indicate the absence of binding, the absence of retrieval, or the absence of both. As a result we cannot interpret the absence of an interaction to indicate that the corresponding features were not bound. Recent research strongly suggests that what can be affected by situational, attentional, and personal factors is mainly retrieval, supporting the conjecture that binding may be pretty automatic (Zmigrod, de Sonneville, Colzato, Swaab, & Hommel, 2013).

3. Experiment 1

3.1. Age attended and race unattended

In this experiment, age served as the relevant social category, whereas race, location, and response were the irrelevant attributes. Twenty five participants were recruited to this experiment. Fig. 3 depicts the interaction patterns observed. Table 1 presents the ANOVA results. As can be seen, a main effect of Race [$F(1, 24) = 10.4$, $\eta_p^2 = 0.30$, $p < 0.005$] showed that repetitions of race yielded faster RTs (793 ms) than alterations of race (831 ms). Most importantly, evidence for the creation and retrieval of "face files" was provided by the emergence of several significant two way interactions. First, the Age × Race interaction was significant [$F(1, 24) = 19.28$, $p < 0.00005$, $\eta_p^2 = 0.34$], signaling the integration of age and race. The following significant differences supported the presence of a binding process by this interaction. For age repetitions, RTs were faster when race repeated (771 ms) than when race alternated (849 ms) [$t(24) = 3.83$, $p < 0.00005$, $d = 0.26$]. For race repetitions, faster RTs were recorded when age also repeated (771 ms) than when age alternated (815 ms) [$t(24) = 4.90$, $p = 0.000001$, $d = 0.15$]. For race alterations, faster RTs obtained when age also altered (849 ms) than when age repeated (813 ms) [$t(24) = 2.74$, $p < 0.01$, $d = 0.12$]. It should be emphasized that the interpretation of the binding effect requires focusing on the interaction as such, and the main effects, whether significant or insignificant, are irrelevant to the interpretation of the binding effect (Hommel, 1998).

The Age × Location interaction was also significant [$F(1, 24) = 4.28$, $p < 0.05$, $\eta_p^2 = 0.51$], attesting to the creation of an age-location binding. The following significant differences supported the presence of a binding process. For age repetitions, faster RTs obtained when location repeated (801 ms) than when location alternated (818 ms) [$t(24) = 2.05$, $p < 0.05$, $d = 0.06$]. For location repetitions, a marginally significant effect showed that RTs were faster when age repeated (801 ms) than when age alternated (812 ms) [$t(24) = 1.46$, $p = 0.07$, $d = 0.04$]. The Race x Response was also significant, [$F(1, 24) = 11.6$, $p < 0.005$, $\eta_p^2 = 0.32$], signaling the binding of the task-irrelevant category of race and response. For response repetitions, faster responses were recorded when race repeated (776 ms) than when race alternated (839 ms) [$t(24) = 5.07$, $p < 0.000001$, $d = 0.20$]. The Location × Response interaction was marginally significant [$F(1, 24) = 3.84$, $p = 0.06$, $\eta_p^2 = 0.13$], such that when response repeated, RTs were faster for location repetition (797 ms) than for location alteration (818 ms) [$t(24) = 2.59$, $p < 0.01$, $d = 0.06$]. All the other three- and two-way interactions were not significant.

4. Experiment 2

4.1. Age attended and sex unattended

In this experiment, age served as the relevant dimension, whereas sex, location, and response were the irrelevant dimensions. Seventeen participants were recruited for this Experiment. Fig. 4 presents the mean RTs patterns observed in this experiment. Table 2 gives the ANOVA results. The interaction of Age × Response was significant [$F(1, 16) = 17.83$, $p < 0.005$, $\eta_p^2 = 0.52$], attesting to the binding of age with response. For age repetitions, a marginally significant effect was documented in which response repetitions yielded faster (912 ms) RTs than response alternations (954 ms) [$t(16) = 1.51$, $p = 0.07$, $d = 0.15$]. For response repetitions, age repetitions produced faster RTs (912 ms) than age alterations (948 ms) [$t(16) = 2.51$, $p = 0.01$, $d = 0.15$], and for response alternations, RTs were faster when age alternated too (915 ms) than when age repeated (945 ms) [$t(16) = 3.24$, $p < 0.005$, $d = 0.14$]. The Age × Sex interaction was significant [$F(1, 16) = 7.93$, $p < 0.05$, $\eta_p^2 = 0.33$], attesting to the binding of the social categories of age and sex. For sex repetitions, faster RTs were produced when age repeated (909 ms) than when age alternated (931 ms) [$t(16) = 1.75$, $p < 0.05$, $d = 0.1$]; and for age alterations, marginally significant faster RTs obtained when age alternated too (932 ms)

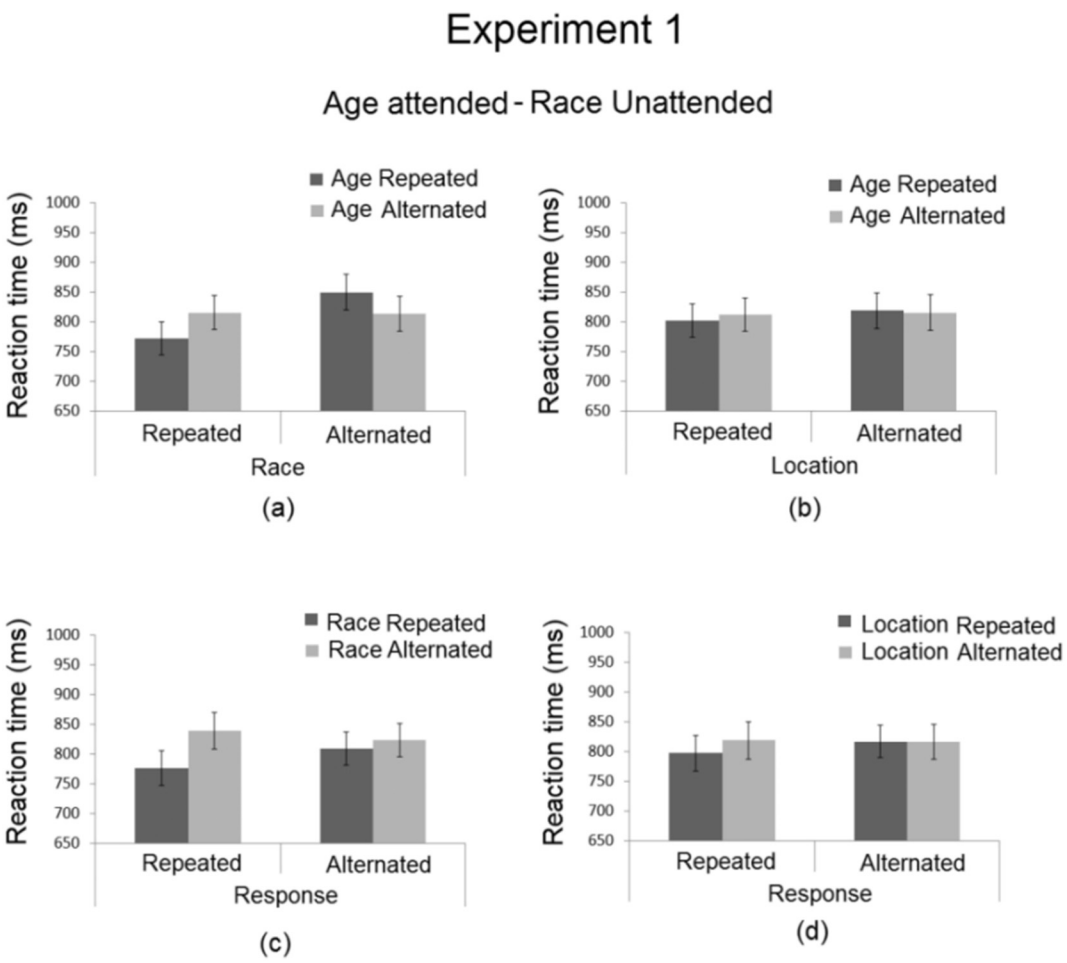


Fig. 3. Binding effects in Experiment 1: (a) age and race, (b) age and location, (c) race and response, and (d) location and response.

than when age repeated (948 ms) [t (16) = 1.58, p = 0.06, d = 0.07]. When age repeated, sex repetitions produced faster RTs (909 ms) than sex alterations (948 ms) [t (16) = 2.58, p < 0.01, d = 0.17]. Finally, the Sex × Response was significant [F (1, 16) = 10.2, p < 0.05, η_p^2 = 0.38], suggesting that the task-irrelevant category of sex, was bound with response. For response repetitions, faster RTs were produced when sex repeated (911 ms) than when sex alternated (949 ms) [t (16) = 3.69, p < 0.001, d = 0.16]. No other significant effects were found.

5. Experiment 3

5.1. Sex-attended and age-unattended

In this experiment, sex served as the relevant social category, whereas age, location, and response location were the irrelevant categories. Fifteen participants were recruited to this Experiment. Fig. 5 presents the mean RTs observed in this experiment. Table 3 presents the ANOVA results. The Sex × Location interaction was significant

Table 1
ANOVA results of Experiment 1.

Effect	df	MSE	F	p	η_p^2
R (Response)	1, 24	7911	0.26		0.01
A (Age)	1, 24	1715	0.40		0.01
C (Race)	1, 24	145,675	10.47	***	0.30
Location (L)	1, 24	10,196	1.93		0.07
R × A	1, 24	38,322	1.74		0.06
R × C	1, 24	59,037	11.6	***	0.32
R × L	1, 24	10,523	3.84		0.13
A × C	1, 24	159,916	19.28	***	0.44
A × L	1, 24	5343	4.28	*	0.15
C × L	1, 24	10	0		0
A × C × L	1, 24	3709	1.81		0.07
R × A × C	1, 24	9882	1.10		0.04
R × A × L	1, 24	2113	0.79		0.03
R × C × L	1, 24	3093	1.10		0.04
R × A × C × L	1, 24	91	0.01		0

* = p < 0.05, ** = p < 0.01, *** = p < 0.00.

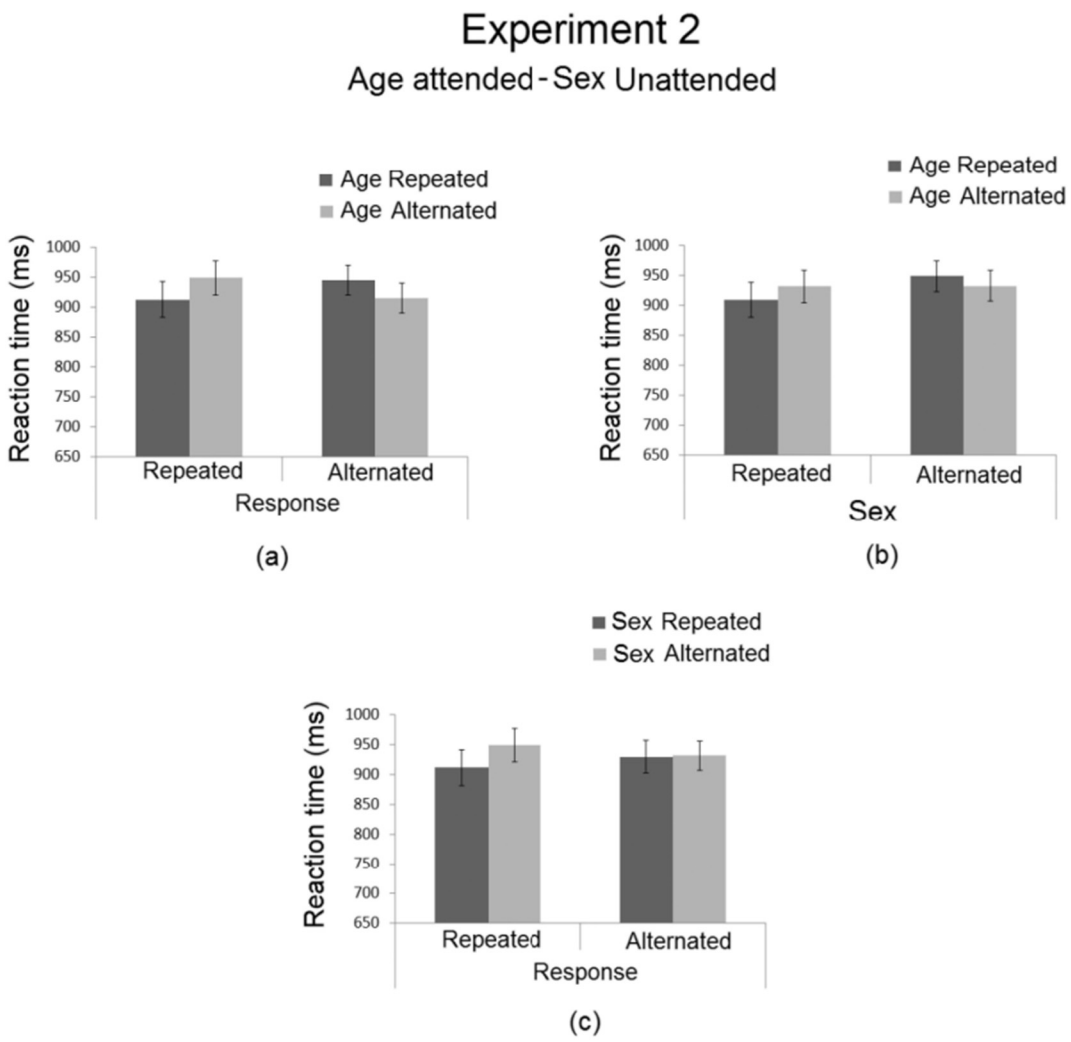


Fig. 4. Binding effects in Experiment 2: (a) age and response, (b) age and sex, and (c) sex and response.

[F (1, 14) = 4.97, p < 0.05, $\eta_p^2 = 0.26$], indicating the binding of the sex with location. The following differences were found to be significant in this interaction, supporting a binding process. When sex repeated, faster RTs obtained when location also repeated (721 ms) than when location alternated (743 ms) [t (14) = 2.28, p < 0.05, d = 0.14]; for location repetitions, a marginally significant effect showed that RTs were faster when sex repeated (721 ms) than when sex alternated (733 ms) [t (14) = 1.69, p = 0.064, d = 0.08]. The Sex × Response was significant too [F (1, 14) = 6.38, p < 0.05, $\eta_p^2 = 0.31$], indicating the binding of sex with response. The following differences gave currency to the notion of binding in this interaction. For sex repetition, a marginally significant effect documented faster RTs in response repetition trials (720 ms) than in response alternation trials (743 ms) [t(14) = 1.60, p = 0.06, d = 0.15]; for response repetitions, RTs were faster

Table 2
ANOVA results of Experiment 2.

Effect	df	MSE	F	p	η_p^2
R (Response)	1, 16	2	0		0
A (Age)	1, 16	668	0.11		0
S (Sex)	1, 16	27,732	3.19		0.16
Location (L)	1, 16	3137	0.74		0.04
R × A	1, 16	75,035	17.83	***	0.52
R × S	1, 16	23,161	10.2	***	0.38
R × L	1, 16	3504	0.28		0
A × S	1, 16	25,123	7.92	*	0.33
A × L	1, 16	853	0.21		0.01
S × L	1, 16	1174	0.39		0.02
A × S × L	1, 16	1225	0.32		0.01
R × A × S	1, 16	7053	1.49		0.08
R × A × L	1, 16	2190	0.42		0
R × S × L	1, 16	784	0.09		0
R × A × C × L	1, 16	4632	1.81		0

* = p < 0.05, ** = p < 0.01, *** = p < 0.005.



Fig. 5. Binding effects in Experiment 3: (a) sex and location, and (b) sex and response.

when sex repeated (720 ms) than when sex alternated (747 ms) [t (14) = 2.48, p < 0.05, d = 0.20], and for response alternations, a marginally significant difference showed that RTs were faster when sex also alternated (718 ms) compared to trials in which sex repeated (743 ms) [t(14) = 1.68, p = 0.05, d = 0.17]. The Sex × Age interaction was far from significance [F (1, 14) = 0.65, p = 0.43, $\eta_p^2 = 0.04$]. This can occur either because the social categories of age and were not bound, or because they were bound but not retrieved (see for Zmigrod et al., 2013). All the other effects were far from significance (all Fs < 1).

6. Experiment 4

6.1. Race-attended and sex-unattended

In this experiment, race served as the relevant social category, whereas sex, location, and response were the irrelevant dimensions. Fourteen participants were recruited to this experiment.

Fig. 6 depicts the mean RTs pattern observed in this experiment. Table 4 gives the ANOVA results. A main effect of Sex [F (1, 13) = 7.64, p < 0.05, $\eta_p^2 = 0.37$] entailed that repetitions of this social category produced faster classification speed (866 ms) than alterations (890 ms). Most importantly, the Race x Response interaction was significant [F (1, 13) = 5.71, p < 0.05, $\eta_p^2 = 0.30$], indicating the binding of the task-relevant social category of race and response. Comparisons showed that for race repetitions, faster RTs obtained when response repeated (870 ms) than when response alternated (901 ms) [t (13) = 2.67, p < 0.01, d = 0.21]; for response repetitions, and for response alterations, faster RTs obtained when race alternated (867 ms) than repeated (901 ms) [t(13) = 3.46, p < 0.005, d = 0.24].

The Race x Sex interaction was significant [F (1, 13) = 5.96, p < 0.05, $\eta_p^2 = 0.31$], indicating the binding of race and sex. The following significant comparison supported binding. For race repetitions, RTs were faster when sex repeated (866 ms) than when sex alternated (905 ms) [t (13) = 3.35, p < 0.005, d = 0.27]. The Sex × Response interaction was significant [F (1, 13) = 17.84, p < 0.001, $\eta_p^2 = 0.57$], indicating the binding of sex with the location of response. Further analyses of this interaction showed that for sex repetitions, faster RTs obtained when response repeated (846 ms) than alternated (884 ms) [t(13) = 3.09, p < 0.005, d = 0.27]; for response repetitions, faster RTs were recorded when sex repeated (846 ms) compared to when sex alternated (895 ms) [t(13) = 6.94, d = 0.37, p < 0.0000001]. The location x response interaction was marginally significant [F (1, 13) = 4.23, p = 0.060, $\eta_p^2 = 0.24$]. Further analyses of this interaction showed that for location repetition, faster RTs obtained when response repeated (866 ms) than when response alternated (891 ms) [t (13) = 2.36, p < 0.05, d = 0.19], and for response alteration, faster RTs obtained when location altered too (877 ms) than repeated (891 ms), although the difference was only marginally significant [t (13) = 1.42, p = 0.08, d = 0.1].

Table 3
ANOVA results of Experiment 3.

Effect	df	MSE	F	p	η_p^2
R (Response)	1, 14	613	0.1		0
A (Age)	1, 14	98	0.08		0
S (Sex)	1, 14	4	0		0
Location (L)	1, 14	5790	1.15		0.07
R × A	1, 14	776	0.12		0
R × S	1, 14	40,524	6.38	*	0.31
R × L	1, 14	4945	1.85		0.11
A × S	1, 14	4858	0.65		0
A × L	1, 14	1142	0.98		0
S × L	1, 14	8473	4.97	*	0.26
A × S × L	1, 14	122	0.1		0
R × A × S	1, 14	1674	1.33		0.08
R × A × L	1, 14	39	0.01		0
R × S × L	1, 14	1329	0.39		0
R × A × S × L	1, 14	92	0.03		0

* = p < 0.05, ** = p < 0.01, *** = p < 0.005.

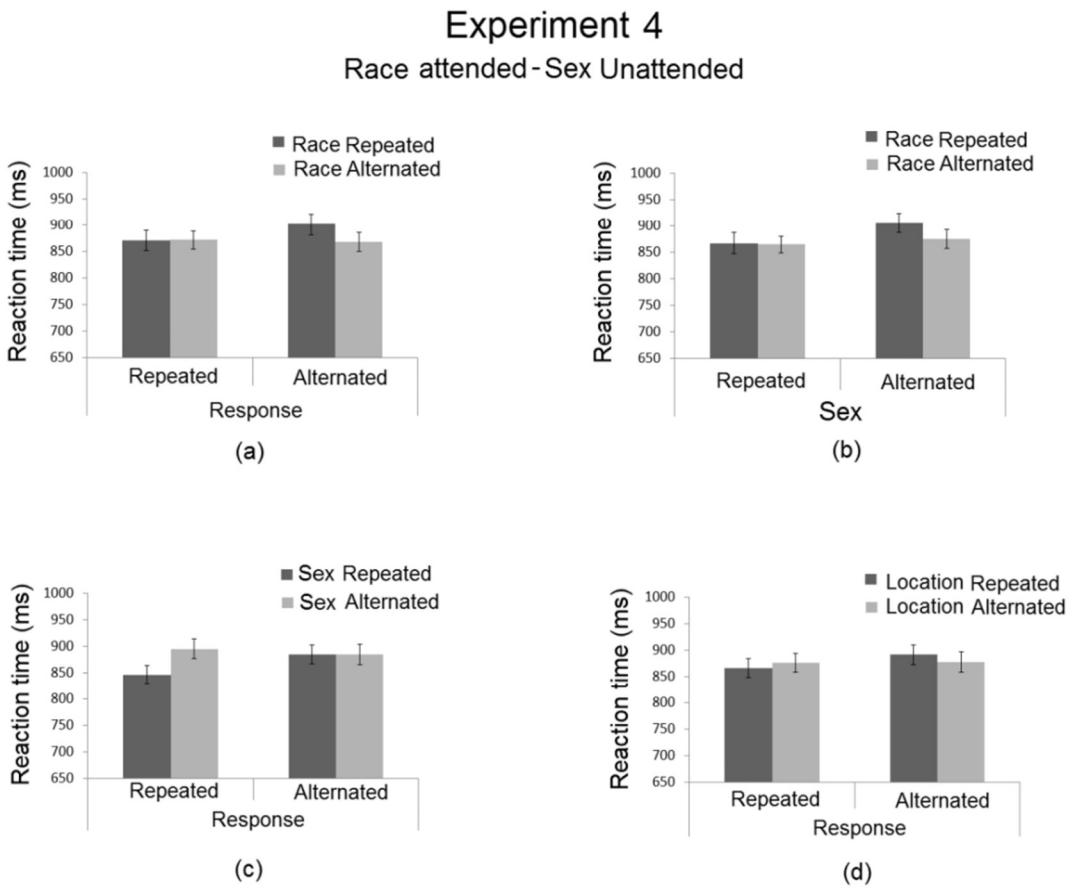


Fig. 6. Binding effects in Experiment 4: (a) race and response, (b) race and sex, (c) sex and response, and (d) location and response.

7. Experiment 5

7.1. Sex-attended and race-unattended

In this experiment, sex served as the relevant social categories, whereas race, location, and response were the irrelevant categories. Thirteen participants were recruited to this experiment. Fig. 7 presents the mean RTs pattern observed in this experiment. Table 5 gives the ANOVA results. A main effect of response [$F(1, 12) = 12.02$, $p < 0.005$, $\eta_p^2 = 0.50$], documented faster RTs for alternations of response (798 ms) than for repetitions (832 ms). A main effect of sex [$F(1, 12) = 14.64$, $p < 0.005$, $\eta_p^2 = 0.54$], entailed faster RTs for sex repetitions (802 ms) than for sex alternations (828 ms). A main effect of race [$F(1, 12) = 4.85$, $p < 0.05$, $\eta_p^2 = 0.28$], suggested faster RTs for race repetitions (809 ms) than for race alternations (821 ms). Most importantly, the Sex x Response interaction was significant [$F(1, 12) = 13.39$, $p < 0.005$, $\eta_p^2 = 0.52$], suggesting the binding of the task-relevant social category with response. Further analyses of this interaction supported the notion of binding. For response repetitions, sex repetitions were responded to faster (799 ms) than sex alterations (863 ms) [$t(12) = 4.89$, $p < 0.0005$, $d = 0.29$]; for sex alterations, faster RTs were recorded when response also altered (792 ms) than when repeated (863 ms) [$t(12) = 5.12$, $p < 0.00005$, $d = 0.34$].

The interaction of Sex × Race was also significant [$F(1, 12) = 5.41$, $p < 0.05$, $\eta_p^2 = 0.31$]. The following significant differences supported the notion of binding between sex and race. When sex repeated, faster

Table 4
ANOVA results of Experiment 4.

Effect	df	MSE	F	p	η_p^2
R (Response)	1, 13	9544	1.32		0.09
Race (C)	1, 13	62315,	2.42		0.15
S (Sex)	1, 13	33,958	7.64	*	0
Location (L)	1, 13	287	0.06		0
R × S	1, 13	32,807	17.84	***	0.94
R × C	1, 13	17,504	5.71	*	0.30
R × L	1, 13	7788	4.23		0.24
S × C	1, 13	12,083	5.96	*	0.31
S × L	1, 13	17,504	5.71	*	0.30
C × L	1, 13	6752	3.29		0.20
S × C × L	1, 13	5899	3.36		0.20
R × S × C	1, 13	386	0.18		0
R × S × L	1, 13	151	0.13		0
R × C × L	1, 13	11,766	3.61		0.21
R × S × C × L	1, 13	33	0.01		0

* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.005$.

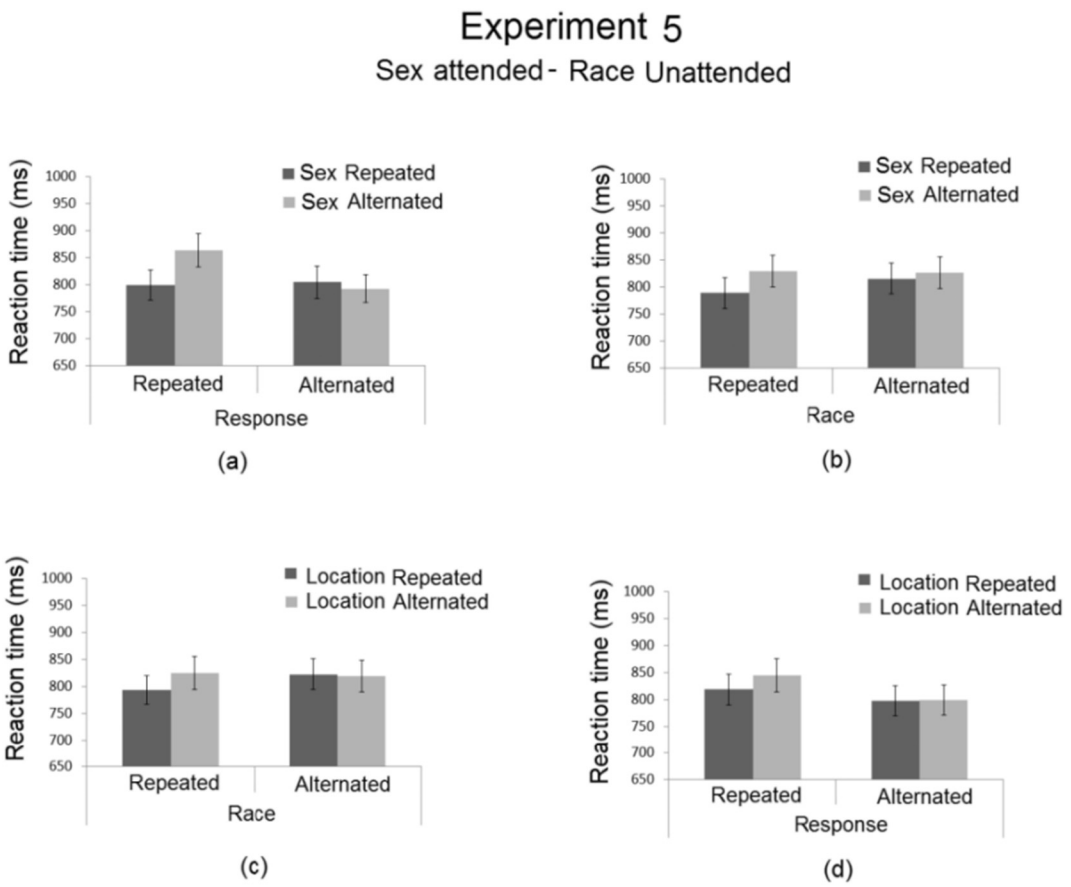


Fig. 7. Binding effects in Experiment 5: (a) sex and response, (b) race and sex, (c) race and location, and (d) location and response.

RTs were recorded for repetition of race (788 ms) compared to alternations (815 ms) [$t(12) = 2.85$, $p < 0.005$, $d = 0.12$]; similarly, when race repeated, faster RTs recorded when sex repeated (788 ms) compared to when it altered (829 ms) [$t(12) = 3.47$, $p < 0.005$, $d = 0.20$]. The Race × Location interaction was significant too [$F(1, 12) = 6.25$, $p < 0.05$, $\eta_p^2 = 0.34$], attesting to the binding of race with spatial location. Further analyses of this interaction showed that, for race repetitions, faster RTs obtained when location repeated (793 ms) than when location alternated (824 ms) [$t(12) = 2.49$, $p < 0.05$, $d = 0.14$]; similarly, for location repetition, faster RTs obtained when race repeated (793 ms) than when race alternated (822 ms) [$t(12) = 3.85$, $p < 0.005$, $d = 0.14$]. Finally, location interacted with response [$F(1, 12) = 5.08$, $p < 0.05$, $\eta_p^2 = 0.29$]. The following significant differences supported the presence of a binding process for these attributes. For repetitions of response, faster RTs obtained when location repeated (818 ms) compared to when it alternated (844 ms) [$t(12) = 2.22$, $p < 0.05$, $d = 0.12$]. For location alterations, faster RTs obtained for response alternations (799 ms) than for repetitions (844 ms) [$t(12) = 3.70$, $p < 0.005$, $d = 0.2$].

8. General discussion

The central question addressed by the current study asked how social categories are extracted from unfamiliar faces. Five experiments

Table 5
ANOVA results of Experiment 5.

Effect	df	MSE	F	p	η_p^2
R (Response)	1, 12	57,922	12.02	***	0.50
Race (C)	1, 12	7758	4.85	*	0.28
S (Sex)	1, 12	34,938	14.64	***	0.54
Location (L)	1, 12	10,795	3.39		0.22
R × S	1, 12	75,799	13.39	***	0.52
R × C	1, 12	7947	5.71	*	0.20
R × L	1, 12	7481	5.08	*	0.29
S × C	1, 12	12,480	5.41	*	0.31
S × L	1, 12	2038	0.34		0.02
C × L	1, 12	14,521	6.25	*	0.34
S × C × L	1, 12	0	0		0
R × S × C	1, 12	386	0.18		0
R × S × L	1, 12	14,017	2.24		0.15
R × C × L	1, 12	10,736	6.10	*	0.83
R × S × C × L	1, 12	10,070	4.03		0.25

* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.005$.

provided substantial evidence for a binding process by which social categories are encoded, maintained, and retrieved together in what we called "face files" structure. Face files are short-term, episodic, binary bindings of facial and/or non-facial (i.e., location, motor) attributes (Fitousi, 2016b). Binding of "face files" was indicated by partial repetition costs (Hommel, 2004), a pattern in which faster RTs are observed when two social categories – a relevant and an irrelevant category – are repeated or alternated together compared to cases in which one of the categories is repeated but the other is alternated. Table 6 provides a succinct summary of the relevant and irrelevant social categories tested in the current experiments and their binding structures.

A closer look at the contents of the "face files" documented here reveals several consistent patterns. First, all experiments, except for Experiment 1, gave rise to visuo-motor integrations with the task-relevant social category (see also Fitousi, 2016b). These visuo-motor bindings are pairwise structures in which the response toward a focal social category is integrated and retrieved with the focal social category (Hommel, 1998). A second pattern that emerges in these experiments is the presence of visuo-motor bindings with task-irrelevant social categories. This pattern has been observed in Experiments 1, 2, 4, and 5. It suggests that even when a social category is not in the focus of attention it can be still activated and integrated with the motor action associated with. A third regularity observed here is the presence of visuo-visuo integrations between the focal social category and the irrelevant social category (i.e., age and sex, age and race, sex and race, race and sex). These patterns were present in all experiments except for Experiment 3, suggesting that "face files" integrate task-relevant and task-irrelevant social categories (see also Martin et al., 2015).

The fourth pattern is pertinent to the involvement of spatial location in binding, which was found to integrate with the social categories of age (Experiment 1), response (Experiment 1, 3), sex (Experiment 4), and race (Experiment 5). It is fair to say that this type of integration has not been consistent throughout the five experiments, but it shows that spatial location may play a role in the binding process (see also Fitousi, 2016b). Location binding is probably not mandatory (Hommel, 1998), but it does occur with many of the tested attributes (Fitousi, 2016b). Information regarding the face's location might be of tremendous importance to the perceiver. First, it carries a survival utility as it indicates the relative distance of the viewer from the constructed person. Second, the possibility that social categories are encoded with spatial codes is not farfetched. Recall that in the present paradigm faces could appear either on top or bottom of the screen. Recent studies have shown that categorizations of concepts such as powerful and powerless (Schubert, 2005), or moral and immoral (Meier, Sellbom, & Wygant, 2007) are influenced by their vertical position on the screen (Lakens, 2012). There are good reasons to believe that social categories are associated with various stereotypes of power, morality, and strength. Vertical position effects, as those found by Schubert (2005) and Lakens (2012) might be found with social categories of faces as well.

Another important finding in the current investigation concerns the dominant role of sex in binding. In Fitousi (2016b), it was facial identity that was part of the "face file" whether it was task-relevant or not. In the current series of experiments, the social category of sex was part of the "face file" whether it was task-relevant or not (Experiments 2, 4). Also notable is the asymmetric pattern of binding of sex with age. Interestingly, when sex served as the task-relevant category it was not bound with age, but when age was the focal category and sex served as the task-irrelevant category, it did integrate with age. This asymmetry is somewhat opposite to recent finding in the mouse-tracking paradigm (Cloutier et al., 2014), as well as in the Garner paradigm (Quinn & Macrae, 2005), that found interference of age with sex, but not of sex with age.

All of the patterns observed in the current experiments are commensurate with the notion of "intentional weighting" which has been proposed recently by Hommel, Memelink, Zmigrod, and Colzato (2014) to account for binding processes with low-level features. According to this idea, the chances of a feature, or a social category in the present case, to be integrated into an "face file" depends on its relevancy for a particular task (Hommel, 2004). Facial categories that bear particular survival utility, such as, sex or race of a face are probably given more weight, and thus are more likely to enter a "face file" structure.

One immediate implication of the current results is that social categories are encoded and extracted simultaneously and automatically, even when they are irrelevant for the task at hand. The results are commensurate with the idea that faces are embodied entities (Spivey & Dale, 2004), which activate a rich network of action programs and perceptual codes in their viewers (Freeman & Ambady, 2011). The mere presence of a face generates a series of pairwise episodic bindings that most often consist of the response emitted toward the face, as well as other focal and non-focal primary attributes. The findings of visuo-visuo bindings with social categories of sex, race, and age are most commensurate with the idea that person construal is an automatic process (Allport, 1954; Bargh, 1999), such that the task-relevant category interacts with other task-irrelevant categories. The observation of visuo-motor binding fits nicely with the idea that social categories trigger motor programs, a conjecture supported by event-related potential (ERP) studies showing that the extraction of facial attributes (e.g., sex, race, age) is immediately and concomitantly shared with the motor cortex (Freeman, Ambady, Midgley, & Holcomb, 2011).

8.1. Relations to the Martin et al. (2015) study

At the outset I have referred to a study by Martin et al. (2015) who have deployed a similar sequential categorization paradigm to the one used here. These researchers have found improved performance in trials in which the relevant (e.g., sex) and irrelevant (e.g., race) social categories repeated or alternated together from trial *n-1* to trial *n* compared to trials in which only one of the social categories repeated but the other alternated. The current results replicate and extend these outcomes in several important ways. First, the current study found similar patterns of *partial repetition costs* with social categories, and thus adduced further support to the notion of simultaneous extraction of social categories (Fitousi & Wenger, 2013; Freeman & Ambady, 2011; Quinn & Macrae, 2005). Second, using the priming setting the current study showed that response and stimulus binding effects can be disentangled into visuo-motor and visuo-visuo integrations. It is likely that the binding effects observed by Martin et al. (2015) conflated these two types of integration mechanisms. Third, although the design used by Martin et al. (2015) afforded binding between three social categories at once (i.e., sex + age + race) such triple binding effects have not been found. The three-way interactions in the Martin et al. study were not

Table 6
Summary of binding structures found in Experiments 1–5. In all the experiments response location and spatial location were irrelevant attributes.

Experiment	Relevant category	Irrelevant category	Binding structures
1	Age	Race	Age + Race, Race + Response, Age + Location
2	Age	Sex	Age + Response, Sex + Response, Age + Sex
3	Sex	Age	Sex + Response, Sex + Location
4	Race	Sex	Race + Response, Race + Sex, Sex + Response, Sex + Location
5	Sex	Race	Sex + Response, Sex + Race, Race + Location Race + Response, Response + Location

significant. This suggested that each social category influenced performance independently of the other, a result that is fully consistent with the binary bindings found here and advocated by the face file approach (Fitousi, 2016b). Fourth, with the current method I could demonstrate the involvement of spatial location in binding. Although, location has not been found to be critical for the creation of a "face file", it did play a role in binding social categories, maybe through the coding of social status (Lakens, 2012). Finally, the study by Martin et al. was not conducted with the a-priori goal of investigating binding processes. The method they have used was borrowed from the bilingualism literature (Thomas & Allport, 2000), while the priming method deployed here has been specifically developed to address binding processes (Hommel, 1998, 2004).

8.2. Convergence and binding as complementary mechanisms

A prominent theory by Freeman and Ambady (2011) proposes that perception of social categories in a face is a dynamic process that evolves over hundreds of milliseconds. The final representation in this process is shaped by interactive top-down and bottom up influences. The model is instantiated in an attractor dynamic network (Churchland & Sejnowski, 1989), in which social attributes, stereotypes, and task demands are represented as nodes in an interconnected network. Depending on its initial condition, the network gravitates toward a certain attractor state, which represents its final status. The network models the integration of multiple sensory, motor, and cognitive information sources across hundreds of millisecond until a final representation of the perceived face is reached. One implication of Freeman and Ambady's (2011) model is that perceptual processing of irrelevant social categories (e.g., sex), can partially activate, and therefore interfere, with other social categories (e.g., race) as well as with the motor actions associated with them. The model by Freeman and Ambady (2011) addresses a large scale of social phenomena at various processing levels, taking into consideration top-down and bottom-up influences. The model excels at explaining the cross-categories interactions (Cloutier et al., 2014), doing so by adhering to a mechanism of convergence of evidence from multiple (neuronal) sources. The binding effects proposed here can be viewed as reflecting a complementary mechanism to the convergence mechanism proposed by Freeman and Ambady (2011). That is, a fast, automatic, episodic binding process that operates in parallel to the dynamic convergence process.

8.3. Person perception: categorical perception or low-level featural processing?

One of the most critical aspects in the study of face construal concerns the degree to which person perception is held categorically (Allport, 1954). By categorical perception researchers often mean that perceivers have abstracted the low-level features of the face to come up with a higher level representation. This newly created representation is general enough to make contact with long-term knowledge such as membership of the face owner in a social group (e.g., Old, Black). However, researchers often find it difficult to make inferences regarding such categorical processes with absolute confidence (Martin et al., 2015). There is always the possibility that participants engaged in categorization tasks, such as classifying the sex of faces, have not represented the faces categorically as "MALE" vs. "FEMALE", but simply reacted according to some differentiating low-level set of features (e.g., the width of the jaw). Using a large sample of faces might not necessarily fix this. Participants might be still operating on the low-level features. Such a criticism can be leveled against the current study. The small number of faces[2] used may have led to configural learning of low-level features rather than the genuine integration of complex facial attributes. Although this possibility might be plausible it is very unlikely. Five lines of evidence speak against such a possibility. First, in case that people respond to the configuration, the $2 \times 2 \times 2$ stimulus-feature design should have partitioned into two conditions; in one condition S2 is the exact replica of S1, and in the second condition it is not. This should have resulted in a four-way interaction, in which only exact repetitions speed up response repetitions and slow down response alterations, while all seven other conditions do the opposite. Since the current findings do not seem to reflect such a case, they provide positive evidence against configurational processing. Second, previous studies on feature binding found the effects of binding and learning to be independent of each other (Colzato, Raffone, & Hommel, 2006; Hommel & Colzato, 2009). Third, the similarity between the current results and those of Martin et al. (2015) who have used a larger set of faces provides another source of confidence that the effects do not reflect configural learning. Fourth, there is substantial evidence that facial attributes such as emotion, race, and gender are perceived categorically (Etcoff & Magee, 1992; Beals & Keil, 1995; Calder, Young, Perrett, Etcoff, & Rowland, 1996), and there is good reason to believe that the effects documented here operate at a categorical rather a featural level.

Finally, the finding of asymmetric binding patterns strongly suggests that participants were not attending to low-level features but rather constructed high-level categorical representations. In particular, it was found that age was bound with sex (Experiment 2), but that sex was not integrated with age (Experiment 3). Note that the stimulus set in these two experiments was identical, such that the low-level features of the faces in the two categorization tasks were the same. However, when participants categorized the faces on age they were influenced by changes in the featural cues of sex, but when they classified the faces on sex they were not affected by changes in the featural cues of age. A similar pattern has been found in a recent study by Fitousi (2016b) with the social dimensions of emotion and gaze direction. If participants based their categorizations only on low level features, chances are high that symmetric binding patterns should have emerged. Asymmetric patterns of interference have been also recorded in the Garner paradigm (Quinn & Macrae, 2005, see also Fitousi & Wenger, 2013), and a similar argument has been raised by Quinn and Macrae (2005) to account for this asymmetry.

8.4. Face files: possible extensions and implications for social psychology phenomena

The emerging representations demonstrated in the current study are identity representations. As such, they may contain other non-visual categorical information such as the social status of the face or their relations to the self (Hommel & Colzato, 2015). Extending these representations to include other non-visual features such as moral values, valence, or trust, may allow researchers to explain various social phenomena. At a first look it is difficult to see how a cognitive model of binding between low-level stimulus features (Hommel, 2004), or even between mid-level face attributes (Fitousi, 2016a, 2016b) can be used to explain relatively complex social effects. However, this might be possible if one appreciates the fact that many of the observed phenomena in social psychology can be traced back to domain-general mechanisms (Tversky & Kahneman, 1983).

In the subsequent sections, I will demonstrate the utility of such a binding approach. It should be noted that I am not the first one to do so. Hommel and his colleagues (Kim & Hommel, 2015, see also Hommel & Colzato, 2015) have explained conformity and trust by adhering to binding mechanisms. These authors have argued that: "the principles and concepts… were originally not developed to account for social phenomena. This implies that at least some aspects and phenomena of social conformity emerge from domain-general characteristics of the human cognitive system" (Kim & Hommel, 2015, p. 489). These authors have also highlighted the existing similarity between

[2] There is nothing, in principle, that precludes the use of a larger number of faces in the Hommel's (1998) paradigm employed here. However, this should come on the expense of a tight experimental control over specific stimuli repetitions and alterations.

the binding account and the theory developed by Greenwald et al. (2002) to address implicit attitudes in social cognition.

8.5. Implicit association test and binding

The implicit association test (IAT, Greenwald, McGhee, & Schwartz, 1998) allows researchers to measure automatic attitudes toward social groups. This feature has turned the IAT into one of the most popular paradigms in social psychology. To see how the paradigm works, consider the following example offered by Greenwald et al. (1998) in their seminal paper. In one experimental block Black versus White faces are classified into two groups. In a second block, pleasant versus unpleasant words are classified into two groups. In a third block, classification is extended to include exemplars from both categories (i.e. faces and words). Now, the exact pairing of the face's race and word pleasantness turns out to be critical. If classification under the Black + unpleasant vs. White + pleasant assignment is facilitated compared to classification under the Black + pleasant vs. White + unpleasant assignment, then one can infer the involvement of preexisting associations between the face's race and the polarity of attitude toward its group.

The emergence of associations between faces and social attitudes in the IAT paradigm can be readily accounted for by the "face files" approach advanced here. When presented with a face, people automatically create a face file that contains binding structures of task relevant, task irrelevant, and motor/action codes. According to this idea, incongruent classification (Black + pleasant vs. White + Unpleasant) are more difficult to classify than congruent configurations (Black + unpleasant vs. White + pleasant), because the former necessitate the retrieval of two conflicting binding structures – a long-term structure that is based on persistent attitudes and a new structure that relies on short-term binding. Partial overlap between the two emerging representations leads to *partial repetition costs*. In congruent conditions, the newly created binding structure is in full agreement with the long-term binding structure, and consequently partial repetition costs are minimized. The "face file" approach (Hommel, 1998; Fitousi, 2016b) can thus be used to further test the relative contributions of short-term and long-term processes in generating implicit social attitudes. For example, by presenting Black or White faces alongside pleasant or unpleasant words, one can test whether *partial repetition costs* occur, and whether the strength of these binding effects depends on the congruency or incongruency of the exact stereotypic configurations. Moreover, response/or action binding can be dissociated from stimulus feature binding.

8.6. Social status, spatial metaphors, and binding

Living in a social group requires members to recognize and represent the social hierarchy of themselves and others. This knowledge is important in order to make correct inferences regarding own and others social roles during group interactions (Ridgeway & Diekema, 1989). It is often the case that high-ranked individuals are posited at higher spatial positions than low-ranked individuals. The association between social status and spatial position is more than a metaphor (Lakoff & Johnson, 1980). It is highly likely that the physical aspects of social status are embodied in the human mind (Schubert, 2005; see also Lakens, Semin, & Foroni, 2011). One way of studying this type of embodiment is by presenting words (Lakens, 2012), or faces (Dahl & Adachi, 2013) of social import in either a spatially-congruent or spatially-incongruent locations on the computer screen. In one such study, Dahl and Adachi (2013) examined whether social status metaphors exist in monkeys. They have presented the faces of high- and low-status chimpanzees one above the other, and found that performance was faster when the social status of the face was *congruent* with its location in space (e.g., high-ranked individual presented above a low-ranked individual) compared to an *incongruent* condition (e.g., high-ranked individual presented below a low-ranked individual). These results can be readily explained by the current "face files" approach. According to this interpretation, chimpanzees[3] hold in their brain an internal long-term structure of faces. This structure contains a facial identity that is bound with a related spatial code of its social status (top, bottom). When a new short-term binding "face file" is created in an incongruent condition, it is only in partial overlap with the long-term structure (i.e., same identity but different location code). This leads to the emergence of partial repetition costs. These are minimized when the short-term "face file" structure fully overlaps with the long-term representation. It has been shown in this and in a previous study (Fitousi, 2016b), that spatial location plays an important role in the formation of "face files", especially when the task related aspects refer to some form of spatial codes (e.g., gaze direction). To the best of my knowledge, no other social psychology theory ascribes spatial location a prominent role in social cognition, albeit its importance in the embodiment of social metaphors (Schubert, 2005).

8.7. Trust and binding

Developing trust in other social members of the group is an important social ability. Trust is best defined as: "believing that when offered the chance, he or she is not likely to behave in a way that is damaging to us" (Gambetta, 1988, p. 219). Hommel and Colzato (2015) presented a binding account of interpersonal trust in which the self and others' perceptual features as well as their action-related features are perceived as two subsequent "event files." Hommel and Colzato's basic idea is that an agent represents herself with respect to a set of features (e.g., female, smart, student), and the Other with respect to another set of features (e.g. male, smart, father). The level of trust between oneself and the Other depends on three factors: (a) the degree of overlap in the constituent features. The more common features between the Self and Other, the greater the trust felt toward this person, (b) the relative importance of features (Hommel et al., 2014), and (c) the person's cognitive style of decision making (i.e., inclusive or exclusive). This binding notion and its extension to faces (Fitousi, 2016a, 2016b), gets support from the finding that facial similarity between trustor and trustee stimulates attributions of trustworthiness (DeBruine, 2005; Bailenson, Iyengar, Yee, & Collins, 2009). It is likely that people maintain a "face file" of themselves. This "face file" consists of unique bindings of their physical appearance with their non-visual characteristics (e.g., smart, good). When a person perceives another, he or she retrieves two "face files" – one belongs to herself, and the belongs to the person in whom she should or should not trust. Recall that "face files" consist also of responses and action tendencies. Low degree of overlap between oneself "face file" and the Other person "face file" can result in response conflict, which in turn may elicit feelings of uncertainty and distrust (Hommel & Colzato, 2015).

9. Conclusions

The present work suggests a domain general mechanism of short-term binding among facial attributes, their location in space, and the motor actions emitted toward them. The emerging representations are identity–specific and as such may include many aspects of social import. Although face files are short-term representations, they might easily conflate with long-term representations to produce central social phenomena, such as trust.

Compliance with ethical standards

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Conflict of interest: Daniel Fitousi declares that he has no conflict of interest.

[3] To the best of my knowledge, no such study has been conducted with human faces. It is now underway in my lab.

Ethical approval: All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. Informed consent: Informed consent was obtained from all individual participants included in the study.

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References

Algom, D., & Fitousi, D. (2016). Half a century of research on Garner interference and the separability-integrality distinction. *Psychological Bulletin*, *142*, 1352–1383.
Allport, G. W. (1954). *The nature of prejudice*. MA: Addison-Wesley.
Bailenson, J. N., Iyengar, S., Yee, N., & Collins, N. A. (2009). Facial similarity between voters and candidates causes influence. *Public Opinion Quarterly* nfn064.
Bargh, J. A. (1999). The cognitive monster: The case against controllability of automatic stereotype effects. In S. Chaiken, & Y. Trope (Eds.), *Dual process theories in social psychology* (pp. 361–382).
Bausell, R. B., & Li, Y. F. (2002). *Power analysis for experimental research: A practical guide for the biological, medical and social sciences*. Cambridge University Press.
Beals, J. M., & Keil, F. C. (1995). Categorical effects in the perception of faces. *Cognition*, *57*, 217–239.
Bestelmeyer, P. E., Jones, B. C., DeBruine, L. M., Little, A. C., Perrett, D. I., Schneider, A., ... Conway, C. A. (2008). Sex-contingent face aftereffects depend on perceptual category rather than structural encoding. *Cognition*, *107*(1), 353–365.
Brebner, J. L., Martin, D., & Macrae, C. N. (2009). Dude looks like a lady: Exploring the malleability of person categorization. *European Journal of Social Psychology*, *39*(6), 1109–1119.
Breiter, H. C., Etcoff, N. L., Whalen, P. J., Kennedy, W. A., Rauch, S. L., Buckner, R. L., ... Rosen, B. R. (1996). Response and habituation of the human amygdala during visual processing of facial expression. *Neuron*, *17*(5), 875–887.
Brewer, M. B. (1988). A dual process model of impression formation. In R. S. J. Wyer, & T. K. Srull (Eds.), *Advances in social cognition* (pp. 1–36). Hillsdale.
Bruce, V., & Young, A. (1986). Understanding face recognition. *British Journal of Psychology*, *77*(3), 305–327.
Burton, A. M., Bruce, V., & Johnston, R. A. (1990). Understanding face recognition with an interactive activation model. *British Journal of Psychology*, *81*(3), 361–380.
Burton, A. M., Bruce, V., & Dench, N. (1993). What's the difference between men and women? Evidence from facial measurement. *Perception*, *22*(2), 153–176.
Calder, A. J., & Young, A. W. (2005). Understanding the recognition of facial identity and facial expression. *Nature Reviews Neuroscience*, *6*(8), 641–651.
Calder, A. J., Young, A. W., Perrett, D. I., Etcoff, N. L., & Rowland, D. (1996). Categorical perception of morphed facial expressions. *Visual Cognition*, *3*(2), 81–118.
Churchland, P. S., & Sejnowski, T. J. (1989). Neural representation and neural computation. In L. Nadel, L. A. Cooper, P. Culicover, & R. M. Harnish (Eds.), *Neural connections, mental computation* (pp. 15–48). Cambridge, MA: MIT Press.
Cloutier, J., Freeman, J. B., & Ambady, N. (2014). Investigating the early stages of person perception: The asymmetry of social categorization by sex vs. age. *PloS One*, *9*(1), e84677.
Colzato, L. S., Raffone, A., & Hommel, B. (2006). What do we learn from binding features? Evidence for multilevel feature integration. *Journal of Experimental Psychology: Human Perception and Performance*, *32*, 705–716.
Dahl, C. D., & Adachi, I. (2013). Conceptual metaphorical mapping in chimpanzees (*Pan troglodytes*). *eLife*, *2*, e00932.
DeBruine, L. M. (2005). Trustworthy but not lust-worthy: Context-specific effects of facial resemblance. *Proceedings of the Royal Society of London B: Biological Sciences*, *272*(1566), 919–922.
Devine, P. G. (1989). Stereotypes and prejudice: Their automatic and controlled components. *Journal of Personality and Social Psychology*, *56*(1), 5–18.
Dishon-Berkovits, M., & Algom, D. (2000). The Stroop effect: It is not the robust phenomenon that you have thought it to be. *Memory & Cognition*, *28*(8), 1437–1449.
Enlow, D. H., & Moyers, R. E. (1982). *Handbook of facial growth*. WB Saunders Company.
Etcoff, N. L., & Magee, J. J. (1992). Categorical perception of facial expressions. *Cognition*, *44*, 227–240.
Farah, M. J., Wilson, K. D., Drain, M., & Tanaka, J. N. (1998). What is "special" about face perception? *Psychological Review*, *105*(3), 482–498.
Felleman, D. J., & Van Essen, D. C. (1991). Distributed hierarchical processing in the primate cerebral cortex. *Cerebral Cortex*, *1*(1), 1–47.
Fiske, S. T., & Neuberg, S. L. (1990). A continuum of impression formation, from category—based to individuating processes: Influences of information and motivation on attention and interpretation. *Advances in Experimental Social Psychology*, *23*, 1–74.
Fitousi, D. (2013). Mutual information, perceptual independence, and holistic face perception. *Attention, Perception, & Psychophysics*, *75*, 983–1000.
Fitousi, D. (2015). Composite faces are not processed holistically: Evidence from the Garner and redundant target paradigms. *Attention, Perception & Psychophysics*, *77*, 2037–2060.
Fitousi, D. (2016a). Comparing the role of selective and divided attention in the composite face effect: Insights from Attention Operating Characteristic (AOC) plots and cross-contingency correlations. *Cognition*, *148*, 34–46.
Fitousi, D. (2016b). What's in a face file? Feature binding with facial identity, emotion, and gaze direction. *Psychological Research* (in press).
Fitousi, D., & Wenger, M. J. (2013). Variants of independence in the perception of facial identity and expression. *Journal of Experimental Psychology: Human Perception and Performance*, *39*, 133–155.
Freedman, D. J., Riesenhuber, M., Poggio, T., & Miller, E. K. (2001). Categorical representation of visual stimuli in the primate prefrontal cortex. *Science*, *291*(5502), 312–316.
Freeman, J. B., & Ambady, N. (2011). A dynamic interactive theory of person construal. *Psychological Review*, *118*(2), 247–279.
Freeman, J. B., Ambady, N., Midgley, K. J., & Holcomb, P. J. (2011). The real-time link between person perception and action: Brain potential evidence for dynamic continuity. *Social Neuroscience*, *6*(2), 139–155.
Freeman, J. B., Nakayama, K., & Ambady, N. (2013). Finger in flight reveals parallel categorization across multiple social dimensions. *Social Cognition*, *31*(6), 792–805.
Gambetta, D. (1988). *Trust: Making and breaking cooperative relations*.
Garner, W. R. (1974). *The processing of information and structure*. Potomac, MD: Erlbaum.
Greenwald, A. G., McGhee, D. E., & Schwartz, J. L. (1998). Measuring individual differences in implicit cognition: The implicit association test. *Journal of Personality and Social Psychology*, *74*(6), 1464–1480.
Greenwald, A. G., Banaji, M. R., Rudman, L. A., Farnham, S. D., Nosek, B. A., & Mellott, D. S. (2002). A unified theory of implicit attitudes, stereotypes, self-esteem, and self-concept. *Psychological Review*, *109*(1), 3–25.
Haxby, J. V., Hoffman, E. A., & Gobbini, M. I. (2000). The distributed human neural system for face perception. *Trends in Cognitive Sciences*, *4*(6), 223–233.
Hommel, B. (1998). Event files: Evidence for automatic integration of stimulus–response episodes. *Visual Cognition*, *5*, 183–216.
Hommel, B. (2004). Event files: Feature binding in and across perception and action. *Trends in Cognitive Sciences*, *8*, 494–500.
Hommel, B., & Colzato, L. S. (2009). When an object is more than a binding of its features: Evidence for two mechanisms of visual feature integration. *Visual Cognition*, *17*, 120–140.
Hommel, B., & Colzato, L. S. (2015). Interpersonal trust: an event-based account. *Frontiers in Psychology*, 6.
Hommel, B., Proctor, R. W., & Vu, K. P. L. (2004). A feature-integration account of sequential effects in the Simon task. *Psychological research*, *68*(1), 1–17.
Hommel, B., Memelink, J., Zmigrod, S., & Colzato, L. S. (2014). Attentional control of the creation and retrieval of stimulus–response bindings. *Psychological Research*, *78*(4), 520–538.
Hubel, D. H., & Wiesel, T. N. (1977). Ferrier lecture: Functional architecture of macaque monkey visual cortex. *Proceedings of the Royal Society of London B: Biological Sciences*, *198*(1130), 1–59.
Kahneman, D., & Treisman, A. (1984). Changing views of attention and automaticity. *Varieties of attention. vol. 1*. (pp. 29–61).
Kahneman, D., Treisman, A., & Gibbs, B. J. (1992). The reviewing of object files: Object-specific integration of information. *Cognitive Psychology*, *24*, 174–219.
Kanwisher, N., McDermott, J., & Chun, M. M. (1997). The fusiform face area: A module in human extrastriate cortex specialized for face perception. *The Journal of Neuroscience*, *17*(11), 4302–4311.
Kareev, Y. (1995). Positive bias in the perception of covariation. *Psychological Review*, *102*(3), 490–502.
Keizer, A. W., Colzato, L. S., & Hommel, B. (2008). Integrating faces, houses, motion, and action: Spontaneous binding across ventral and dorsal processing streams. *Acta Psychologica*, *127*(1), 177–185.
Kim, D., & Hommel, B. (2015). An event-based account of conformity. *Psychological Science*, *26*(4), 484–489.
Lakens, D. (2012). Polarity correspondence in metaphor congruency effects: Structural overlap predicts categorization times for bipolar concepts presented in vertical space. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *vol. 38*, 726–736.
Lakens, D., Semin, G. R., & Foroni, F. (2011). Why your highness needs the people. *Social Psychology*, *42*, 205–213.
Lakoff, G., & Johnson, M. (1980). *Metaphors we live by*. University of Chicago press.
Levin, D. T. (2000). Race as a visual feature: Using visual search and perceptual discrimination tasks to understand face categories and the cross-race recognition deficit. *Journal of Experimental Psychology: General*, *129*(4), 559–574.
Livingstone, M. S., & Hubel, D. H. (1987). Psychophysical evidence for separate channels for the perception of form, color, movement, and depth. *Journal of Neuroscience*, *7*(11), 3416–3468.
Livingstone, M., & Hubel, D. (1988). Segregation of form, color, movement, and depth: Anatomy, physiology, and perception. *Science*, *240*(4853), 740–749.
MacLin, O. H., & Malpass, R. S. (2001). Racial categorization of faces: The ambiguous race face effect. *Psychology, Public Policy, and Law*, *7*(1), 98–118.
Macrae, C. N., Bodenhausen, G. V., & Milne, A. B. (1995). The dissection of selection in person perception: Inhibitory processes in social stereotyping. *Journal of Personality and Social Psychology*, *69*(3), 397–407.
Martin, D., Swainson, R., Slessor, G., Hutchison, J., Marosi, D., & Cunningham, S. J. (2015). The simultaneous extraction of multiple social categories from unfamiliar faces. *Journal of Experimental Social Psychology*, *60*, 51–58.
McGurk, H., & MacDonald, J. (1976). Hearing lips and seeing voices. *Nature*, *264*(5588), 746–748.

Meier, B. P., Sellbom, M., & Wygant, D. B. (2007). Failing to take the moral high ground: Psychopathy and the vertical representation of morality. *Personality and Individual Differences*, *43*, 757–767.
Melara, R. D., & Algom, D. (2003). Driven by information: A tectonic theory of Stroop effects. *Psychological Review*, *110*, 422–471.
Quinn, K. A., & Macrae, C. N. (2005). Categorizing others: The dynamics of person construal. *Journal of Personality and Social Psychology*, *88*(3), 467–479.
Ridgeway, C., & Diekema, D. (1989). Dominance and collective hierarchy formation in male and female task groups. *American Sociological Review*, 79–93.
Rogers, T. T., & McClelland, J. L. (2004). *Semantic cognition: A parallel distributed processing approach*. MIT press.
Rolls, E. T., & Tovee, M. J. (1995). Sparseness of the neuronal representation of stimuli in the primate temporal visual cortex. *Journal of Neurophysiology*, *73*(2), 713–726.
Schubert, T. W. (2005). Your highness: Vertical positions as perceptual symbols of power. *Journal of Personality and Social Psychology*, *89*, 1–21.
Singer, W., & Gray, C. M. (1995). Visual feature integration and the temporal correlation hypothesis. *Annual Review of Neuroscience*, *18*(1), 555–586.
Spivey, M. J., & Dale, R. (2004). On the continuity of mind: Toward a dynamical account of cognition. *Psychology of Learning and Motivation*, *45*, 87–142.
Sugase, Y., Yamane, S., Ueno, S., & Kawano, K. (1999). Global and fine information coded by single neurons in the temporal visual cortex. *Nature*, *400*(6747), 869–873.
Thomas, M. S., & Allport, A. (2000). Language switching costs in bilingual visual word recognition. *Journal of Memory and Language*, *43*(1), 44–66.
Treisman, A. (1996). The binding problem. *Current Opinion in Neurobiology*, *6*(2), 171–178.
Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, *12*(1), 97–136.
Treisman, A., & Schmidt, H. (1982). Illusory conjunctions in the perception of objects. *Cognitive Psychology*, *14*(1), 107–141.
Tversky, A., & Kahneman, D. (1983). Extensional versus intuitive reasoning: The conjunction fallacy in probability judgment. *Psychological Review*, *90*(4), 293–315.
Von der Malsburg, C. (1999). The what and why of binding: the modeler's perspective. *Neuron*, *24*(1), 95–104.
Zmigrod, S., de Sonneville, L. M., Colzato, L. S., Swaab, H., & Hommel, B. (2013). Cognitive control of feature bindings: Evidence from children with autistic spectrum disorder. *Psychological Research*, *77*(2), 147–154.